

GPS versus Local Knowledge Teacher Notes

TOK Cognitive Science Activity Bank • Knowledge and Technology • 25 min

Category	Knowledge and Technology	Time	25 min
Difficulty	Easy	ID	gps-versus-local-knowledge

Big idea: Data can be precise and incomplete.

Overview

Students compare precise digital knowledge with situated local knowledge.

Student Prompt

Inspect GPS versus Local Knowledge Stimulus A. Record one first claim, the evidence you used, your confidence, and what would make you revise.

Concrete Stimulus Packet

GPS versus Local Knowledge Stimulus A	Student-facing stimulus 1: A route-planning scenario where the map suggests a faster but impractical route. Rank, sort, or audit this output. Identify which rule, ranking signal, or interface choice shaped your judgement.	Students inspect this exact card first and commit to a claim before hearing anyone else.
GPS versus Local Knowledge Stimulus B	Student-facing stimulus 2: Local knowledge cards: school gate closed, roadworks, unsafe crossing, steep hill. Rank, sort, or audit this output. Identify which rule, ranking signal, or interface choice shaped your judgement.	Use this for comparison, a second round, or a partner challenge.
GPS versus Local Knowledge Reveal / Twist	Student-facing stimulus 3: A decision table balancing speed, safety, reliability, and lived experience. Rank, sort, or audit this output. Identify which rule, ranking signal, or interface choice shaped your judgement.	Teacher reveal: connect the changed response to the big idea — Data can be precise and incomplete.

Actual Lesson Questions and Key

#	Question for students	Teacher key / cue
1	After inspecting GPS versus Local Knowledge Stimulus A, what is your first knowledge claim?	Teacher cue: look for a clear claim, not just a description.
2	Which exact detail from GPS versus Local Knowledge Stimulus A did you use as evidence?	Teacher cue: students should name visible words, data, source features, method features, or contextual clues.
3	What did you infer beyond the evidence?	Teacher cue: this is where assumptions, framing, models, or perspective usually appear.
4	How confident are you: 50%, 60%, 70%, 80%, 90%, or 100%?	Teacher cue: confidence is data for the debrief, not a grade.
5	What missing information would most improve the knowledge claim?	Teacher cue: push for method, source, context, comparison, or definitions.
6	When is local knowledge better than technological knowledge?	Teacher cue: students should move from the classroom example to a general TOK issue.

Teacher Key / Reveal

The key move is not the “right answer”; it is the moment students see how data can be precise and incomplete.

- Strong response: separates observation from inference.
- Strong response: names a method, source, model, language choice, context, or perspective.
- Strong response: revises confidence when new evidence or a new criterion appears.
- Weak response: gives only opinion or says “it depends” without criteria.

Setup Checklist

- Prepare a starter stimulus using: A route-planning scenario where the map suggests a faster but impractical route.
- Print or project the student response grid before the activity begins.
- Plan a private first-response moment before any group discussion.

Ready-to-Use Examples

- A route-planning scenario where the map suggests a faster but impractical route.
- Local knowledge cards: school gate closed, roadworks, unsafe crossing, steep hill.
- A decision table balancing speed, safety, reliability, and lived experience.

Suggested Timing

Stage	Teacher move	Watch for
1	Give students a route recommended by an app.	Assumptions, confidence shifts, evidence claims, and language choices.
2	Add local knowledge: roadworks, locked gate, dangerous crossing, staircase, informal shortcut, crowding at certain times.	Assumptions, confidence shifts, evidence claims, and language choices.
3	Groups design the best route for different users.	Assumptions, confidence shifts, evidence claims, and language choices.
4	Compare algorithmic optimisation with human judgement.	Assumptions, confidence shifts, evidence claims, and language choices.
Debrief	Move from the activity result to a TOK knowledge question.	Students distinguishing method, evidence, interpretation, and overclaiming.

Teacher Language

- Write your first judgement before we discuss it; the first answer is useful evidence.
- Separate what you noticed from what you inferred.
- What would make this knowledge claim more reliable?

Common Misconceptions

- Students may treat a tool's output as neutral because it feels automatic.
- Students may focus only on user error and miss design incentives or hidden rules.

Assessment Look-Fors

- Identifies the rule, design choice, or incentive shaping knowledge.
- Distinguishes access to information from justified knowledge.

Differentiation and Adaptations

- Short lesson: use only the first example, one response grid, and one debrief question.
- Long lesson: add a second round where students revise the method and compare outcomes.
- Low-tech option: run with printed cards, sticky notes, and a board tally.

Extension Tasks

- Design a second version of the activity that tests the same knowledge issue in another AOK.
- Find a real-world example where the same knowledge problem matters outside the classroom.
- Write a 150-word TOK exhibition-style commentary using the activity as the object context.

Related Activities

- Algorithmic Feed Simulation
- Search Engine Ranking Game
- AI Hallucination Courtroom